QUESTION ANALYSIS

EEE339

Power Electronics

Question Analysis

Sessions 2020–21 *1 papers, 31 questions, 59 formulas*

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1. At a glance:

This analysis covers **1 paper** (2020-21), with 31 questions and 190 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

With one paper, no trend can be read. Use this as a map of the topics.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Power devices	48	25 %	8	Diodes (16); Thyristors (12); TRIAC and BJT (8)
Rectifiers	42	22 %	6	Three-phase (19); Full-wave (13); Half-wave (10)
DC-DC converters	34	18 %	6	Buck (15.5); Boost (11); SEPIC and Cuk (5)
Inverters	27	14 %	4	PWM (10); Filters (7); Multilevel (6)
Power computations	19	10 %	3	Distortion (13); Nonlinear loads (6)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2020-21	Term tests	All
Power computations	**13**	**6**	**19**
Distortion	**13**	·	13
Nonlinear loads	·	**6**	6
Power devices	**39**	**9**	**48**
Diodes	**12**	**4**	16
Thyristors	**7**	**5**	12
TRIAC and BJT	**8**	·	8
IGBT	**7**	·	7
Comparison	**5**	·	5
Rectifiers	**32**	**10**	**42**
Three-phase	**14**	**5**	19
Full-wave	**8**	**5**	13
Half-wave	**10**	·	10
AC control and cycloconverters	**5**	·	**5**
Cycloconverters	**5**	·	5
DC-DC converters	**19**	**15**	**34**
Buck	**8**	**7.5**	15.5
Boost	**6**	**5**	11
SEPIC and Cuk	**5**	·	5
Field chopping	·	**2.5**	2.5
DC power supplies	**6**	·	**6**
Linear regulator	**6**	·	6
Inverters	**27**	·	**27**
PWM	**10**	·	10
Filters	**7**	·	7
Multilevel	**6**	·	6
Square-wave	**4**	·	4

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Topic and subtopic	2020-21	Term tests	All
Applications	**9**	·	**9**
Solar PV	**5**	·	5
HVDC	**4**	·	4

3. Priority list:

Must know: Power devices (48), Rectifiers (42), DC-DC converters (34), Inverters (27), Power computations (19), Applications (9), DC power supplies (6), AC control and cycloconverters (5)

Likely: none

Less often: none

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. Power computations:

19 marks, 3 questions, in 1 of 1 papers. Book: Hart Ch. 2.

Most often numerical (2 of 3). Subtopics: Distortion; Nonlinear loads.

Formulas:

Subtopic	Formula	Equation	Used in
Distortion (13)	Rms of a distorted wave	${I_{rms}}^2 = \sum_n {I_n}^2$	20-21 1(a)
	Distortion factor	$DF = \dfrac{I_1}{I_{rms}}$	20-21 1(a)
	Total harmonic distortion	$THD = \dfrac{\sqrt{I_{rms}^2 - I_1^2}}{I_1}$	20-21 1(a)
	DF and THD	$DF = \sqrt{\dfrac{1}{1+THD^2}}$	20-21 1(a)
	Power factor	$pf = \dfrac{P}{S} = DF\cos\theta_1$	20-21 1(a)
	One harmonic only	$I_n = THD\, I_1$	20-21 1(d)
Nonlinear loads (6)	Crest factor	$CF = \dfrac{I_{peak}}{I_{rms}}$	20-21 7(d)
	Power with a sine source	$P = V_1 I_1 \cos\theta_1$	20-21 7(d)

Symbols:

I_{rms} = rms of the whole wave (A) ↩

I_n = rms value of harmonic n (A) ↩

DF = distortion factor ↩

I_1 = rms fundamental (A) ↩

THD = total harmonic distortion ↩

pf = power factor ↩

P = average power (W) ↩

S = apparent power, $V_{rms}I_{rms}$ (VA) ↩

θ_1 = angle of the fundamental current ↩

CF = crest factor ↩

I_{peak} = peak of the wave (A) ↩

V_1 = rms source voltage (V) ↩

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Questions:

No.	Paper	Marks	Type	Question
Distortion				
1	20-21 1(a)	7	derivation	Define distortion factor (DF) and total harmonic distortion (THD). Derive that $DF = \sqrt{\frac{1}{1+(THD)^2}}$
2	20-21 1(d)	6	numerical	A sinusoidal current of 10A rms at a 60 Hz fundamental frequency is contaminated with a ninth harmonic current. The current is expressed as $i(t) = 10\sqrt{2}\sin(2\pi 60t) + I_9\sqrt{2}\sin(18\pi 60t)$ A. Determine the value of the ninth harmonic rms current I9 if the THD is -i) 5 percent ii) 20 percent.
Nonlinear loads				
3	20-21 TT1 7(d)	6	numerical	A sinusoidal voltage source of $v(t) = 240\sqrt{2}\sin(2\pi 60t)$ V is applied to a nonlinear load, resulting in a nonsinusoidal current which is expressed in Fourier series form as $i(t) = 8\sin(2\pi 60t) + 4\sin(4\pi 60t)$ A. Determine (a) the power absorbed by the load, (b) the power factor of the load, (c) the distortion factor of the load current, (d) the total harmonic distortion of the load current, (e) crest factor of the load current.

4.2. Power devices:

48 marks, 8 questions, in 1 of 1 papers. Book: Rashid Ch. 2, 4, 9.

Most often theory (6 of 8). Subtopics: Diodes; Thyristors; IGBT; TRIAC and BJT; Comparison.

Formulas:

Subtopic	Formula	Equation	Used in
Diodes (16)	Recovery time	$t_{rr} = t_a + t_b$	20-21 1(b)
	Peak reverse current	$I_{RR} = t_a \frac{di}{dt}$	20-21 1(b)
	Storage charge	$Q_{RR} \approx \frac{1}{2}\frac{di}{dt}t_{rr}^2$	20-21 1(c)
	Peak from the charge	$I_{RR} = \sqrt{2Q_{RR}\frac{di}{dt}}$	20-21 1(c)
Thyristors (12)	**Two-transistor model**	$I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1-(\alpha_1+\alpha_2)}$	20-21 2(a)
	Turn-on condition	$\alpha_1 + \alpha_2 \approx 1$	20-21 2(a)

Symbols:

t_{rr} = reverse recovery time (s) ↩

t_a = time to the reverse peak (s) ↩

t_b = fall time after the peak (s) ↩

I_{RR} = peak reverse current (A) ↩

di/dt = rate of fall of current (A/s)

Q_{RR} = stored charge removed during recovery (C) ↩

I_A = anode current (A) ↩

α_1, α_2 = common-base gains of the two transistors ↩

I_G = gate current (A) ↩

I_{CBO1}, I_{CBO2} = leakage currents of the two transistors (A) ↩

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Questions:

No.	Paper	Marks	Type	Question
Diodes				
1	20-21 1(b)	6	theory	Define reverse recovery current and reverse recovery time for diode switching. What is the importance of these parameters?
2	20-21 1(c)	6	numerical	The manufacturer of a selected diode gives the rate of fall of the diode current di/dt=20 A/μs, and its reserve *[sic: should be reverse]* recovery time t_{rr}=5 μs. What value of peak reverse current and the storage charge do you expect?
3	20-21 TT1 7(a)	4	theory	Explain the reverse recovery time of diode with diagram.
Thyristors				
4	20-21 2(a)	7	derivation	Draw the two transistors model of a thyristor and explain the turn-on condition of a thyristor with equations from this model.
5	20-21 TT1 7(b)	5	theory	Draw the I-V curve of thyristor. Define Holding current and Latching current for thyristor
IGBT				
6	20-21 3(b)	7	theory	Draw the cross section of an IGBT showing the layers. Differentiate between punch-through and non-punch through IGBTs.
TRIAC and BJT				
7	20-21 6(c)	8	theory	Explain the structure of TRIAC with diagram. Mention the constructional feature of a Power BJT.
Comparison				
8	20-21 6(d)	5	theory	Compare the following device in terms of power capacity and switching speed: BJT, MOSFET, GTO, IGBT, MCT

4.3. Rectifiers:

42 marks, 6 questions, in 1 of 1 papers. Book: Hart Ch. 3, 4.

Most often numerical (4 of 6). Subtopics: Half-wave; Full-wave; Three-phase.

Formulas:

Subtopic	Formula	Equation	Used in
Half-wave (10)	Impedance and angle	$Z = \sqrt{R^2 + (\omega L)^2},\ \theta = \tan^{-1}\dfrac{\omega L}{R}$	20-21 2(b)
	RL load current	$i = \dfrac{V_m}{Z}\sin(\omega t - \theta) + Ae^{-\omega t/\omega\tau}$	20-21 2(b)
	Constant	$A = -\dfrac{V_m}{Z}\sin(\alpha - \theta)e^{\alpha/\omega\tau}$	20-21 2(b)
	Average, half-wave	$I_o = \dfrac{1}{2\pi}\displaystyle\int_\alpha^\beta i\, d(\omega t)$	20-21 2(b)
	Rms, half-wave	${I_{rms}}^2 = \dfrac{1}{2\pi}\displaystyle\int_\alpha^\beta i^2\, d(\omega t)$	20-21 2(b)
	Load power	$P = I_{rms}^2 R$	20-21 2(b)
	Half-wave output	$V_o = \dfrac{V_m}{2\pi}(\cos\alpha - \cos\beta)$	20-21 2(b)
Full-wave (13)	**Controlled bridge, R load**	$V_o = \dfrac{V_m}{\pi}(1 + \cos\alpha)$	20-21 2(c)

Subtopic	Formula	Equation	Used in
	Rms, controlled bridge	$V_{rms} = \frac{V_m}{\sqrt{2}}\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}}$	20-21 2(c)
	Diode bridge	$V_o = \frac{2V_m}{\pi}$	20-21 7(c)
Three-phase (19)	**Diode bridge**	$V_o = \frac{3V_m}{\pi} = 0.955V_m$	20-21 3(a)
	Controlled bridge	$V_o = \frac{3V_m}{\pi}\cos\alpha$	20-21 3(d)
	Harmonics (controlled)	$V_n = \frac{3V_m}{\pi}\frac{2}{n^2-1}\sqrt{\cos^2\alpha + n^2\sin^2\alpha}$	20-21 3(d)
	Harmonics (diode)	$V_n = \frac{2V_o}{n^2-1}$	20-21 8(d)
	Harmonic current	$I_n = \frac{V_n}{\sqrt{R^2 + (n\omega L)^2}}$	20-21 3(d)
	Rms load current	$I_{rms}^2 = I_o^2 + \sum_n \left(\frac{I_n}{\sqrt{2}}\right)^2$	20-21 8(d)
	Rms diode current	$I_{D,rms} = \frac{I_{rms}}{\sqrt{3}}$	20-21 8(d)
	Line current	$i_a = i_{D1} - i_{D4}$	20-21 3(a)

Symbols:

Z = load impedance (Ω) ↩

R = load resistance (Ω) ↩

L = load inductance (H) ↩

ω = angular frequency of the source (rad/s) ↩

θ = load angle (rad) ↩

i = load current (A) ↩

V_m = peak source voltage (V); peak line-to-line voltage (V) ↩

t = time (s) ↩

A = constant set by $i(\alpha) = 0$ (A) ↩

α = delay angle (rad) ↩

I_o = average load current, V_o/R (A) ↩

β = extinction angle, $i(\beta) = 0$ ↩

I_{rms} = rms load current (A) ↩

P = load power (W) ↩

V_o = average output voltage (V) ↩

V_{rms} = rms output voltage (V) ↩

V_n = peak voltage of harmonic n (V) ↩

n = harmonic order (6, 12, 18, ...) ↩

I_n = peak current of harmonic n (A) ↩

$I_{D,rms}$ = rms diode current (A) ↩

i_a = line current of phase a (A) ↩

i_{D1}, i_{D4} = currents of the two diodes on phase a (A) ↩

Questions:

No.	Paper	Marks	Type	Question
Half-wave				
1	20-21 2(b)	10	numerical	For the circuit of Fig. 2(b), the source is 120 V rms at 60 Hz, R=20 Ohm, L=0.04 H, and the delay angle is 45°. Determine (i) an expression for i(ωt) , (ii) the average current, (iii) the power absorbed by the load, and (iv) the power factor.
Full-wave				
2	20-21 2(c)	8	numerical	A full-wave controlled bridge rectifier has an ac input of 120 V rms at 60 Hz and a 20 Ω load resistor. The delay angle is 40°. Determine the average current of the load, the power absorbed by the load, the apparent power of the sourced *[sic: should be source]* and the input pf.

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No.	Paper	Marks	Type	Question
3	20-21 TT1 7(c)	5	diagram	With proper alignment, show the waveshape of output voltage, output current, diode currents and source current for a full wave bridge rectifier with sinusoidal source and R-L load.
Three-phase				
4	20-21 3(a)	8	theory	Draw the configuration of three phase uncontrolled rectifier. Show the source voltages, output voltage, diode currents and line currents with proper alignment.
5	20-21 3(d)	6	numerical	A three phase controlled rectifier has an input voltage of 480 V rms at 60 Hz. The load of the rectifier is resistor of 10 Ω in series with 50 mH inductance. (i) Determine the delay angle required to produce an average current of 50A in the load (ii) Determine the amplitude of the harmonics n=6 and n=12. For harmonics calculation, Fig. 3(d) can be utilized.
6	20-21 TT2 8(d)	5	numerical	A three phase rectifier is supplied by 480 V rms line to line 60 Hz source. The RL load is 100 Ω resistor in series with 15 mH inductor. Determine the rms load current and rms diode currents.

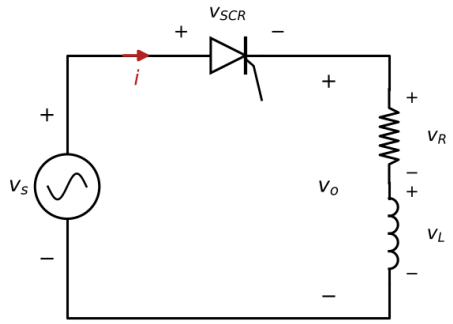


Figure 1: 20-21 2(b): Fig 2(b) of the paper (redrawn)

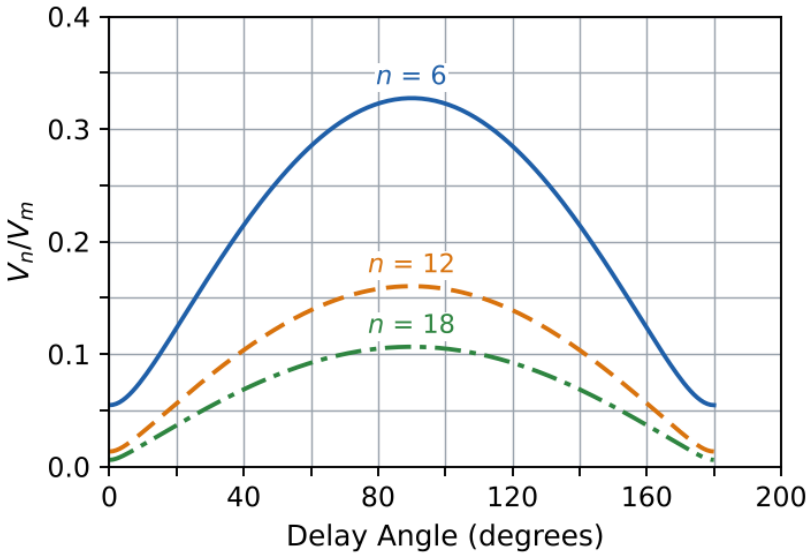


Figure 2: 20-21 3(d): Fig. 3(d) of the paper (redrawn)

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4.4. AC control and cycloconverters:

5 marks, 1 question, in 1 of 1 papers. Book: Hart Ch. 5; Rashid Ch. 11.

Most often theory (1 of 1). Subtopics: Cycloconverters.

Questions:

No.	Paper	Marks	Type	Question
Cycloconverters				
1	20-21 6(b)	5	theory	What is cyclo-converter? Mention its applications.

4.5. DC-DC converters:

34 marks, 6 questions, in 1 of 1 papers. Book: Hart Ch. 6.

Most often theory (2 of 6). Subtopics: Buck; Boost; SEPIC and Cuk; Field chopping.

Formulas:

Subtopic	Formula	Equation	Used in
Buck (15.5)	**Buck output**	$V_o = DV_s$	20-21 4(b)
	Buck ripple current	$\Delta i_L = \frac{V_o(1-D)}{Lf}$	20-21 4(b)
	Diode current	$I_D = (1-D)I_L$	20-21 4(b)
	Switch current	$I_S = DI_L$	20-21 4(b)
	Inductor extremes	$I_{max,min} = I_L \pm \frac{\Delta i_L}{2}$	20-21 8(b)
	Buck ripple voltage	$\frac{\Delta V_o}{V_o} = \frac{1-D}{8LCf^2}$	20-21 8(b)
Boost (11)	**Boost output**	$V_o = \frac{V_s}{1-D}$	20-21 4(c)
	Boost inductor current	$I_L = \frac{V_o^2}{V_s R}$	20-21 4(c)
	Boost ripple current	$\Delta i_L = \frac{V_s D}{Lf}$	20-21 4(c)
	Boost ripple voltage	$\frac{\Delta V_o}{V_o} = \frac{D}{RCf}$	20-21 4(c)
SEPIC and Cuk (5)	SEPIC	$V_o = V_s \frac{D}{1-D}$	20-21 4(d)
	Cuk	$V_o = -V_s \frac{D}{1-D}$	20-21 4(d)
Field chopping (2.5)	Field current	$I_f = \frac{\delta V_s}{R_f}$	20-21 8(c)
	Armature equation	$V_a = E_b + I_a R_a$	20-21 8(c)
	Speed	$\omega = \frac{V_a - I_a R_a}{K_v I_f}$	20-21 8(c)

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Symbols:

D = duty ratio ↩

Δi_L = peak-to-peak inductor ripple current (A) ↩

L = inductance (H)

f = switching frequency (Hz)

I_D = average diode current (A) ↩

I_L = average inductor current (A) ↩

I_S = average switch current (A) ↩

$I_{max,min}$ = largest and smallest inductor current (A) ↩

ΔV_o = peak-to-peak output ripple (V) ↩

C = output capacitance (F)

R = load resistance (Ω)

I_f = field current (A) ↩

δ = chopper duty ratio ↩

V_s = supply voltage (V) ↩

R_f = field resistance (Ω) ↩

V_a = armature voltage (V) ↩

I_a = armature current (A) ↩

R_a = armature resistance (Ω) ↩

E_b = back voltage, $K_v I_f \omega$ (V) ↩

ω = speed (rad/s) ↩

K_v = motor constant ↩

Questions:

No.	Paper	Marks	Type	Question
Buck				
1	20-21 4(b)	8	derivation	Draw the circuit diagram and the waveform of inductor current, diode current and capacitor current for a typical buck converter. Also derive the relation between inductor current and diode current from the waveforms.
2	20-21 TT2 8(b)	7.5	numerical	A buck converter has parameters, Vs=50 V, D=0.4, R=20 Ω, L=400 μH and C=100 μF. The switching frequency is 20 kHz. a) Determine the output voltage, b) Determine the average, minimum and maximum inductor current c) The output ripple voltage.
Boost				
3	20-21 4(c)	6	design	A boost converter has an input of [unclear: 5 V] and an output of 25 W at 15 V. The minimum inductor current must be no less than 50% of the average. The ripple output must be less than 1 percent. The switching frequency is [unclear: 300 kHz]. (i) Determine the duty ratio. (ii) Minimum inductor value (iii) Minimum capacitor value
4	20-21 TT2 8(a)	5	diagram	Draw the circuit diagram, equivalent circuit for switch on and off condition and the waveforms of the inductor current, diode current and load voltage for a typical boost converter.
SEPIC and Cuk				
5	20-21 4(d)	5	theory	Differentiate between SEPIC converter and Ćuk converter.
Field chopping				
6	20-21 TT2 8(c)	2.5	theory	Draw the circuit diagram to control the speed of a separately excited dc motor by chopping field current supply. Explain the operation using required equations.

4.6. DC power supplies:

6 marks, 1 question, in 1 of 1 papers. Book: Hart Ch. 7.

Most often theory (1 of 1). Subtopics: Linear regulator.

Formulas:

Subtopic	Formula	Equation	Used in
Linear regulator (6)	Linear regulator	$V_o = V_{ref}\left(1 + \frac{R_1}{R_2}\right)$	20-21 4(a)
	Regulator loss	$P_{loss} = (V_s - V_o)I_L,\ \eta \approx \frac{V_o}{V_s}$	20-21 4(a)

Symbols:

V_o = output voltage (V) ↩

V_{ref} = reference voltage (V) ↩

R_1, R_2 = divider resistors (Ω) ↩

P_{loss} = power lost in the regulator (W) ↩

V_s = input voltage (V) ↩

I_L = load current (A) ↩

η = efficiency ↩

Questions:

No.	Paper	Marks	Type	Question
Linear regulator				
1	20-21 4(a)	6	theory	Draw a typical circuit of a linear dc voltage regulator. What is the drawback of it? How can it be overcome?

4.7. Inverters:

27 marks, 4 questions, in 1 of 1 papers. Book: Hart Ch. 8.

Most often numerical (2 of 4). Subtopics: Multilevel; PWM; Square-wave; Filters.

Formulas:

Subtopic	Formula	Equation	Used in
Multilevel (6)	Cascade output	$v_o = v_1 + v_2 + v_3$	20-21 5(b)
PWM (10)	Modulation ratios	$m_a = \frac{V_{sine}}{V_{tri}},\ m_f = \frac{f_{carrier}}{f}$	20-21 5(c)
	PWM fundamental	$V_1 = m_a V_{dc}$	20-21 5(c)
	Harmonic current at the carrier	$I_n = \frac{V_n}{n\omega L}$	20-21 5(c)
Square-wave (4)	Square-wave harmonics	$V_n = \frac{4V_{dc}}{n\pi}$	20-21 5(d)
	Square-wave THD	$THD = 48.3\ \%$	20-21 5(d)
Filters (7)	Resonance	$\omega_0 = \frac{1}{\sqrt{LC}}$	20-21 6(a)
	Series filter reactance	$X_n = \omega_0 L\left(n - \frac{1}{n}\right)$	20-21 6(a)
	Load harmonic	$\frac{V_{o,n}}{V_1} = \frac{1}{n}\frac{R}{\sqrt{R^2 + X_n^2}}$	20-21 6(a)
	Capacitor peak	$V_{C,max} \approx \frac{I_1}{\omega_0 C}$	20-21 6(a)

Symbols:

v_o = output voltage (V) ↩

v_1, v_2, v_3 = outputs of the three cells (V) ↩

k = number of cells

m_a = amplitude ratio ↩

V_{sine} = peak of the reference sine (V) ↩

V_{tri} = peak of the carrier triangle (V) ↩

m_f = frequency ratio ↩

$f_{carrier}$ = carrier frequency (Hz) ↩

f = output (fundamental) frequency (Hz) ↩

V_1 = peak fundamental output voltage (V) ↩

I_n = peak current of harmonic n (A) ↩

ω = fundamental angular frequency (rad/s) ↩

L = load inductance (H) ↩

V_n = peak voltage of harmonic n (V) ↩

V_{dc} = dc source (V) ↩

n = harmonic order, odd ↩

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THD = total harmonic distortion ↩

ω_0 = tuned frequency (rad/s) ↩

C = filter capacitance (F) ↩

X_n = filter reactance at harmonic n (Ω) ↩

$V_{o,n}$ = load voltage at harmonic n (V) ↩

R = load resistance (Ω) ↩

$V_{C,max}$ = peak capacitor voltage (V) ↩

I_1 = peak fundamental load current (A) ↩

Questions:

No.	Paper	Marks	Type	Question
Multilevel				
1	20-21 5(b)	6	theory	Draw the circuit of a three-level cascade multi source inverter. Why pattern swapping is important in multi-source inverter circuit?
PWM				
2	20-21 5(c)	10	design	Design a bipolar PWM inverter that will produce a 75-V rms 60-Hz output from a 150 V dc source. The load is a series RL combination with R=12Ω and L=60 mH. Select the switching frequency such that the current THD is less than 10%. Assume the impedance at the carrier frequency is entirely inductive.
Square-wave				
3	20-21 5(d)	4	numerical	Determine the voltage THD of a square wave inverter.
Filters				
4	20-21 6(a)	7	numerical	A 10 Ohm resistive load requires a 1000-Hz, 50-V rms sinusoidal voltage. The THD of the load voltage must be no more than 5 percent. An adjustable dc source is available. (i) Design an inverter for this application. (ii) What is maximum voltage across the capacitor?

Normalized Fourier Coefficients V_n/V_{dc} for Bipolar PWM

	$m_a = 1$	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
$n = 1$	1.00	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10
$n = m_f$	0.60	0.71	0.82	0.92	1.01	1.08	1.15	1.20	1.24	1.27
$n = m_f \pm 2$	0.32	0.27	0.22	0.17	0.13	0.09	0.06	0.03	0.02	0.00

Figure 3: 20-21 5(c): the table printed under Q5(c)

4.8. Applications:

9 marks, 2 questions, in 1 of 1 papers. Book: Rashid Ch. 10, 16; Power Electronics Handbook.

Most often theory (2 of 2). Subtopics: HVDC; Solar PV.

Questions:

No.	Paper	Marks	Type	Question
HVDC				
1	20-21 3(c)	4	theory	Write down the advantages of DC power transmission.
Solar PV				
2	20-21 5(a)	5	theory	Briefly explain the types of switch mode DC-DC converters used in a stand-alone solar PV system.

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5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

No.	Paper	Marks	Subtopic	Points to cover
Power devices				
1	20-21 1(b)	6	Diodes	$t_{rr} = t_a + t_b$: from the current zero crossing until the reverse current dies; I_{RR}: the peak reverse current, set by di/dt and the stored charge. They limit the switching frequency and add current stress, switching loss and voltage spikes; soft and abrupt recovery.
2	20-21 3(b)	7	IGBT	Layers from the collector: p^+ collector, (n^+ buffer in PT), n^- drift, p body, n^+ emitter, gate over the channel. PT: buffer, thin drift, lower on-voltage, poor reverse blocking. NPT: no buffer, thick drift, symmetric blocking, positive temperature coefficient, easy paralleling, rugged.
3	20-21 6(c)	8	TRIAC and BJT	TRIAC: two anti-parallel p-n-p-n structures in one chip with one gate; terminals MT1, MT2, G; conducts both ways, triggered by either gate polarity. Power BJT: vertical n^+-p-n^--n^+, thick lightly doped drift region for blocking, interdigitated emitter, low current gain.
4	20-21 6(d)	5	Comparison	Power capacity: GTO > MCT > IGBT > BJT > MOSFET. Switching speed: MOSFET > IGBT > MCT > BJT > GTO.
5	20-21 TT1 7(a)	4	Diodes	A conducting diode stores minority carriers; when the current reverses they must be removed, so a reverse current flows for $t_{rr} = t_a + t_b$; t_a removes the charge near the junction, t_b the rest; $I_{RR} = t_a\, di/dt$; abrupt and soft recovery.
6	20-21 TT1 7(b)	5	Thyristors	I-V curve: reverse blocking, forward blocking up to V_{BO}, then on with a small drop. Latching current I_L: the minimum anode current just after triggering, with the gate removed, to stay on. Holding current I_H: the minimum current to remain on; below it the thyristor turns off. $I_L > I_H$.
Rectifiers				
7	20-21 3(a)	8	Three-phase	Six-diode bridge; the diode with the highest anode (upper group) and lowest cathode (lower group) conduct; each diode conducts 120°, in pairs for 60°; v_o is the largest line-to-line voltage, $V_o = 0.955 V_{m,L\text{-}L}$; line current $i_a = i_{D1} - i_{D4}$.
8	20-21 TT1 7(c)	5	Full-wave	$v_o = \lvert v_s \rvert$; with a large L the current i_o is nearly level; diodes 1-2 conduct in the positive half, 3-4 in the negative; the source current is $\pm i_o$, nearly a square wave.
AC control and cycloconverters				
9	20-21 6(b)	5	Cycloconverters	A direct ac-to-ac frequency changer with no dc link: two phase-controlled converters (P and N) in anti-parallel, each delay angle varied so the output follows a low-frequency sine. Uses: large low-speed motor drives (mills, kilns), ship propulsion, VSCF aircraft power, traction.
DC-DC converters				
10	20-21 4(d)	5	SEPIC and Cuk	Both have two inductors and a coupling capacitor and give an output above or below the input. SEPIC: positive output, pulsating output current, $V_{C1} = V_s$. Ćuk: inverted output, continuous input and output currents, energy moved through C_1, $V_{C1} = V_s + \lvert V_o \rvert$.
11	20-21 TT2 8(a)	5	Boost	Switch closed: inductor charges from V_s, diode off, capacitor feeds the load. Switch open: inductor in series with the source drives the load through the diode. $V_o = V_s/(1 - D)$.
12	20-21 TT2 8(c)	2.5	Field chopping	A chopper on the field supply sets $I_f = \delta V_s / R_f$; speed $\omega = (V_a - I_a R_a)/(K_v I_f)$, so a lower duty ratio weakens the field and raises the speed (field weakening, above base speed).

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No.	Paper	Marks	Subtopic	Points to cover
DC power supplies				
13	20-21 4(a)	6	Linear regulator	A pass transistor in series with the load, its base driven by an error amplifier comparing a fraction of V_o with a reference. Drawback: the transistor drops $V_s - V_o$ and wastes $(V_s - V_o)I_L$, so $\eta \approx V_o/V_s$ and a big heat sink. Cure: a switched-mode converter, where the transistor is fully on or off.
Inverters				
14	20-21 5(b)	6	Multilevel	Three H-bridges, each with its own dc source, outputs in series, giving levels $0, \pm V, \pm 2V, \pm 3V$; each cell switched at its own angle. Pattern swapping rotates the wide and narrow pulses among the cells each half cycle, so every source delivers the same average power and discharges equally.
Applications				
15	20-21 3(c)	4	HVDC	No reactive power or charging current, so long lines and cables are cheaper beyond the break-even distance; two conductors, lower losses, no skin effect; no stability limit; links asynchronous systems; fast power control.
16	20-21 5(a)	5	Solar PV	The dc-dc converter sits between the array and the battery and sets the array voltage for maximum power point tracking. Buck: array voltage above the battery. Boost: below it. Buck-boost, SEPIC or Ćuk: either way. Each adjusts its duty ratio.

6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Power computations				
1	20-21 1(a)	7	Distortion	Write $i(t)$ as a Fourier series; show $I_{rms}^2 = \sum I_n^2$ (cross terms average to 0); put the harmonics $I_{rms}^2 - I_1^2$ in the THD; square; add 1; use $DF = I_1/I_{rms}$.
Power devices				
2	20-21 2(a)	7	Thyristors	Split p-n-p-n into Q_1 and Q_2; $I_C = \alpha I_E + I_{CBO}$ for each; $I_A = I_{C1} + I_{C2}$ and $I_K = I_A + I_G$; solve for I_A; it grows without limit as $\alpha_1 + \alpha_2 \to 1$.
DC-DC converters				
3	20-21 4(b)	8	Buck	v_L in each interval; straight-line i_L triangle; average capacitor current 0 gives $I_L = I_o$; $i_D = i_L$ only in $(1-D)T$; trapezoid area gives $I_D = (1-D)I_L$ (and $I_S = DI_L$).

7. Design recipes:

Every design asked, in book order. Each recipe gives the steps in the order to take them; the link opens the full working.

No.	Paper	Marks	Design	Recipe
DC-DC converters				
1	20-21 4(c)	6	Boost	$D = 1 - V_s/V_o$; $I_L = P_o/V_s$ from the power balance; $I_{min} = 0.5I_L$ gives Δi_L; $L = V_s DT/\Delta i_L$; $C = D/(Rf\,\Delta V_o/V_o)$.
Inverters				
2	20-21 5(c)	10	PWM	$m_a = V_1/V_{dc}$ from the rms target; fundamental current $I_1 = V_1/\lvert Z_1\rvert$; allowed harmonic current from THD < 10 %; harmonic voltages from the table at m_a; choose m_f so $V_{m_f}/(m_f \omega L)$ meets the limit; $f_{carrier} = m_f \times 60$.

8. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

No.	Paper	Marks	Subtopic	Method	Answer
Power computations					
1	20-21 1(d)	6	Distortion	$THD = I_9/I_1$ with only one harmonic; $I_9 = THD \times I_1$.	0.5 A rms (5 %); 2 A rms (20 %)
2	20-21 TT1 7(d)	6	Nonlinear loads	Only the fundamental carries power: $P = V_{1,rms} I_{1,rms} \cos\theta_1$; $I_{rms} = \sqrt{\sum I_{n,rms}^2}$; $pf = P/(V_{rms} I_{rms})$; $DF = I_1/I_{rms}$; $THD = I_2/I_1$; crest factor = peak/rms.	$P = 1358$ W; $pf = DF = 0.894$; THD = 50 %; CF = 1.64
Power devices					
3	20-21 1(c)	6	Diodes	$Q_{RR} = \frac{1}{2} t_{rr}^2 \, di/dt$ (triangle area, abrupt recovery); $I_{RR} = t_{rr} \, di/dt$.	$I_{RR} = 100$ A; $Q_{RR} = 250$ μC
Rectifiers					
4	20-21 2(b)	10	Half-wave	Forced response $(V_m/Z)\sin(\omega t - \theta)$ plus a decaying exponential fixed by $i(\alpha) = 0$; find the extinction angle β from $i(\beta) = 0$; average and rms current by integration; $P = I_{rms}^2 R$; $pf = P/(V_s I_{rms})$.	$\beta = 216.9°$; $I_o = 2.04$ A; $P = 213$ W; $pf = 0.544$
5	20-21 2(c)	8	Full-wave	$V_o = (V_m/\pi)(1 + \cos\alpha)$, $I_o = V_o/R$; I_{rms} from the rms of the chopped sine; $P = I_{rms}^2 R$; $S = V_s I_{rms}$; $pf = P/S$.	$I_o = 4.77$ A; $P = 673$ W; $S = 696$ VA; $pf = 0.967$
6	20-21 3(d)	6	Three-phase	$V_o = (3V_{m,L\text{-}L}/\pi)\cos\alpha = I_o R$ gives α; read the normalized harmonics V_n/V_o at α from Fig. 3(d); $I_n = V_n/\lvert R + jn\omega L\rvert$.	$\alpha = 39.5°$; $V_6 \approx 144$ V, $V_{12} \approx 70$ V peak; $I_6 = 1.27$ A, $I_{12} = 0.31$ A
7	20-21 TT2 8(d)	5	Three-phase	$V_o = 3V_m/\pi$, $I_o = V_o/R$; 6th and 12th harmonic currents $V_n/\lvert R + jn\omega L\rvert$; I_{rms} = root of the sum of squares; each diode conducts one third of the time, $I_{D,rms} = I_{rms}/\sqrt{3}$.	$I_{rms} = 6.49$ A; $I_{D,rms} = 3.75$ A
DC-DC converters					
8	20-21 TT2 8(b)	7.5	Buck	$V_o = DV_s$; $I_L = V_o/R$; $\Delta i_L = V_o(1-D)/(Lf)$; $I_{max,min} = I_L \pm \Delta i_L/2$; $\Delta V_o = V_o(1-D)/(8LCf^2)$.	$V_o = 20$ V; $I_L = 1$ A, $I_{max} = 1.75$ A, $I_{min} = 0.25$ A; $\Delta V_o = 0.094$ V
Inverters					
9	20-21 5(d)	4	Square-wave	Fourier series of the $\pm V_{dc}$ wave, $b_n = 4V_{dc}/n\pi$; $V_1 = 0.9V_{dc}$ rms; $V_{rms} = V_{dc}$; $THD = \sqrt{V_{rms}^2 - V_1^2}/V_1$.	THD = 48.3 %
10	20-21 6(a)	7	Filters	Square-wave bridge with a series LC filter tuned to 1 kHz; V_{dc} from $V_1 = 0.9V_{dc}$; choose L so the harmonics through the filter give THD at most 5 %; $C = 1/(\omega_0^2 L)$; capacitor peak from the fundamental current times X_C.	$V_{dc} = 55.5$ V; $L = 5$ mH; $C = 5.07$ μF; THD 4.26 %; $V_{C,max} \approx 223$ V

9. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

No.	Paper	Kind	Diagram
Power computations			
1	20-21 1(a)	Waveform	a distorted wave as fundamental plus harmonics
2	20-21 TT1 7(d)	Waveform	current with its fundamental and second harmonic
Power devices			
3	20-21 1(b)	Waveform	reverse recovery current, abrupt and soft
4	20-21 2(a)	Diagram	thyristor layers split into two transistors
5	20-21 2(a)	Circuit	two-transistor model
6	20-21 3(b)	Diagram	IGBT cross sections, punch-through and non-punch-through
7	20-21 6(c)	Diagram	TRIAC layers and symbol
8	20-21 6(c)	Diagram	power BJT vertical structure
9	20-21 6(d)	Graph	power capacity against switching frequency
10	20-21 TT1 7(a)	Diagram	stored carriers in a conducting diode
11	20-21 TT1 7(a)	Waveform	reverse recovery current, abrupt and soft
12	20-21 TT1 7(b)	Graph	thyristor I-V curve
Rectifiers			
13	20-21 2(b)	Waveform	source, output voltage and current of the half-wave controlled rectifier
14	20-21 2(c)	Circuit	full-wave controlled bridge
15	20-21 2(c)	Waveform	source and output voltage
16	20-21 3(a)	Circuit	three-phase diode bridge
17	20-21 3(a)	Waveform	source voltages, output voltage, diode and line currents, aligned
18	20-21 3(d)	Waveform	output of the controlled bridge at the delay angle
19	20-21 3(d)	Graph	reading Fig. 3(d) at the delay angle
20	20-21 TT1 7(c)	Circuit	diode bridge with RL load
21	20-21 TT1 7(c)	Waveform	output voltage and current, diode and source currents, aligned
22	20-21 TT2 8(d)	Circuit	three-phase diode bridge
AC control and cycloconverters			
23	20-21 6(b)	Circuit	single-phase cycloconverter
24	20-21 6(b)	Waveform	input and output
DC-DC converters			
25	20-21 4(b)	Circuit	buck converter
26	20-21 4(b)	Circuit	buck equivalent circuits, switch on and off
27	20-21 4(b)	Waveform	inductor, switch, diode and capacitor currents
28	20-21 4(c)	Circuit	boost converter
29	20-21 4(c)	Waveform	inductor current with the minimum inductor
30	20-21 4(d)	Circuit	SEPIC and Ćuk converters
31	20-21 TT2 8(a)	Circuit	boost converter
32	20-21 TT2 8(a)	Circuit	boost equivalent circuits, switch on and off
33	20-21 TT2 8(a)	Waveform	inductor current, diode current and load voltage

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No.	Paper	Kind	Diagram
34	20-21 TT2 8(b)	Waveform	inductor current of this converter
35	20-21 TT2 8(c)	Circuit	field chopper control of a separately excited motor
36	20-21 TT2 8(c)	Graph	speed against field current
DC power supplies			
37	20-21 4(a)	Circuit	linear regulator with pass transistor and error amplifier
Inverters			
38	20-21 5(b)	Circuit	cascade of three H-bridges
39	20-21 5(b)	Waveform	cell outputs and the stepped total
40	20-21 5(b)	Waveform	pattern swapping
41	20-21 5(c)	Circuit	full-bridge inverter with RL load
42	20-21 5(c)	Waveform	bipolar PWM reference and carrier
43	20-21 5(c)	Spectrum	output harmonics
44	20-21 5(d)	Spectrum	square wave and its harmonics
45	20-21 6(a)	Circuit	inverter with series LC filter
46	20-21 6(a)	Spectrum	bridge and load harmonics
47	20-21 6(a)	Waveform	bridge, load and capacitor voltages
Applications			
48	20-21 3(c)	Block diagram	HVDC link
49	20-21 3(c)	Graph	cost against line length for ac and dc
50	20-21 5(a)	Block diagram	stand-alone PV system
51	20-21 5(a)	Graph	panel current and power against voltage

10. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

Pair	Key difference	Asked in
Power devices		
Punch-through vs non-punch-through IGBT	n^+ buffer, thin drift, low on-voltage, poor reverse blocking vs no buffer, thick drift, symmetric blocking, easy paralleling	20-21 3(b)
BJT vs MOSFET vs GTO vs IGBT vs MCT	power GTO > MCT > IGBT > BJT > MOSFET vs speed MOSFET > IGBT > MCT > BJT > GTO	20-21 6(d)
Latching vs holding current	the least current to stay on just after firing vs the least current to remain on; $I_L > I_H$	20-21 TT1 7(b)
DC-DC converters		
SEPIC vs Ćuk converter	positive output, pulsating output current vs inverted output, continuous output current	20-21 4(d)
DC power supplies		
Linear vs switched-mode regulator	transistor in its active region, $\eta \approx V_o/V_s$ vs transistor on or off, high efficiency, ripple	20-21 4(a)
Applications		
HVDC vs HVAC transmission	converter cost, cheaper lines, no stability limit vs cheap terminals, reactive power, stability limit	20-21 3(c)

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11. Repeated questions:

No repeats among these papers.

12. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical	Diagram	Design
Power computations	·	1	2	·	·
Power devices	6	1	1	·	·
Rectifiers	1	·	4	1	·
AC control and cycloconverters	1	·	·	·	·
DC-DC converters	2	1	1	1	1
DC power supplies	1	·	·	·	·
Inverters	1	·	2	·	1
Applications	2	·	·	·	·

13. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Power devices	48	25 %
2	Rectifiers	42	47 %
3	DC-DC converters	34	65 %
4	Inverters	27	79 %
5	Power computations	19	89 %
6	Applications	9	94 %
7	DC power supplies	6	97 %
8	AC control and cycloconverters	5	100 %

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